WEEK 1 REPORT
Image Captioning using the Flickr8k Dataset

1. Problem Statement

Image captioning is the automatic generation of meaningful natural-language descriptions for images by combining computer vision and natural language processing. The objective of this project is to develop an image captioning system using the Flickr8k dataset. Given an input image, the future model should identify important visual content and generate a clear caption that accurately describes the scene. Week 1 focused on preparing reliable data and a reproducible project structure before model development and training.

2. Dataset Summary

The Flickr8k dataset is a benchmark dataset for image captioning. It contains real-world JPG images, with multiple human-written descriptions for each image. These alternative captions help a model learn that the same visual scene can be described in different but valid ways.

Dataset Feature	Value
Dataset	Flickr8k
Total images	8,091
Total captions	40,455
Captions per image	5
Image format	JPG
Caption file	captions.txt

Exploratory data analysis was performed in Pandas to count images and captions, calculate captions per image, examine caption-length distribution, estimate vocabulary size, display image-caption samples, and check for duplicate, blank, unusually short, or corrupted records. The analysis confirmed 8,091 images and 40,455 captions, with approximately five captions per image. Most captions were around 8-12 words long, and no significant data-quality issues were identified.

3. Week 1 Technical Work

Environment and project setup

The development environment was configured and tested using Pandas, NumPy, Matplotlib, OpenCV, NLTK, spaCy, Hugging Face Transformers and Datasets, torchvision, and Pillow. Small import and test examples were run to confirm that the libraries were installed correctly. A requirements.txt file was created to record project dependencies.

Git and GitHub workflow

A structured Git workflow was practiced, including repository creation, cloning, branch creation, adding files, committing changes, pushing branches, opening Pull Requests, and merging reviewed changes. The Week 1 notebook, preprocessing scripts, dependency file, and project outputs were organized and pushed to GitHub through a Pull Request.

4. Preprocessing Decisions

Text and image preprocessing were completed to create a clean, consistent input pipeline for later deep-learning stages. The main decisions and their purposes are summarized below.

Preprocessing Decision	Purpose
Lowercasing	Treat uppercase and lowercase forms of the same word consistently.
Tokenization	Split each caption into individual words for NLP processing.
Stop-word removal	Reduce common low-information terms during the initial cleaned-text preparation.
Vocabulary building	Create a unique word mapping for later numerical encoding.
Image resizing to 224 x 224	Standardize dimensions for pretrained CNN architectures such as ResNet and VGG.
Image normalization	Scale image values to support stable model convergence.
RGB channel validation	Ensure every image has three channels and is compatible with CNN input requirements.

Text outputs: Cleaned captions were saved to data/processed/captions_clean.csv, while the generated vocabulary was saved to data/processed/vocab.json.

Image outputs: All loaded images were resized, normalized, checked for three-channel RGB consistency, and saved in data/processed/images/.

5. Consolidated Notebook and Review

Week 1 activities were consolidated into a clean notebook containing dataset loading, EDA, representative visualizations, text preprocessing, image preprocessing, validation checks, and saved-output paths. The notebook was organized so that the full workflow can be rerun in sequence.

6. Week 1 Deliverables and Outcome

- Finalized problem statement for image captioning using Flickr8k.
- Completed dataset summary and exploratory data analysis.
- Configured and tested the Python AI/ML environment.
- Completed text and image preprocessing and saved processed outputs.
- Consolidated Week 1 work into a clean, reproducible notebook.
- Created requirements.txt and organized the project directory.
- Pushed all Week 1 code and documentation to GitHub through a Pull Request.
- Completed the Week 1 team check-in/mentor review.

7. Conclusion

Week 1 established the foundation for the image captioning project. The Flickr8k dataset was explored and validated, the development and GitHub environments were prepared, and both captions and images were transformed into organized, standardized outputs. The project is now ready to proceed to feature extraction, sequence preparation, model architecture selection, training, and evaluation in the next phase.